Delayed Job  for

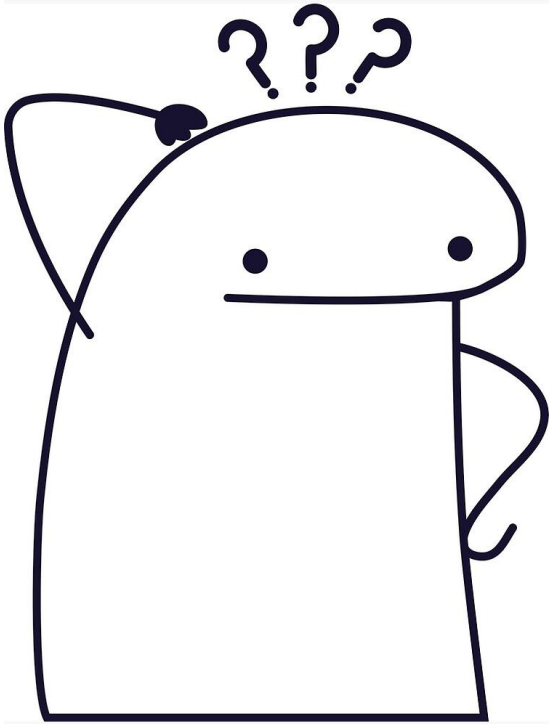
What even is it?

DelayedJob or for short is

- used for asynchronously executing longer tasks in the background
- a direct extraction from shopify where it is used for things like:
  - Sending newsletters
  - Batch imports
  - Spam checks
  - Etc.

How does it work?

- To use it with ActiveRecord we need the delayed_job and the delayed_job_activerecord gems
- After installing the gems we can execute the command rails g delayed_job:active_record
- This generates the migration for the jobs table, which is used by delayed_job to enqueue it’s jobs in the background

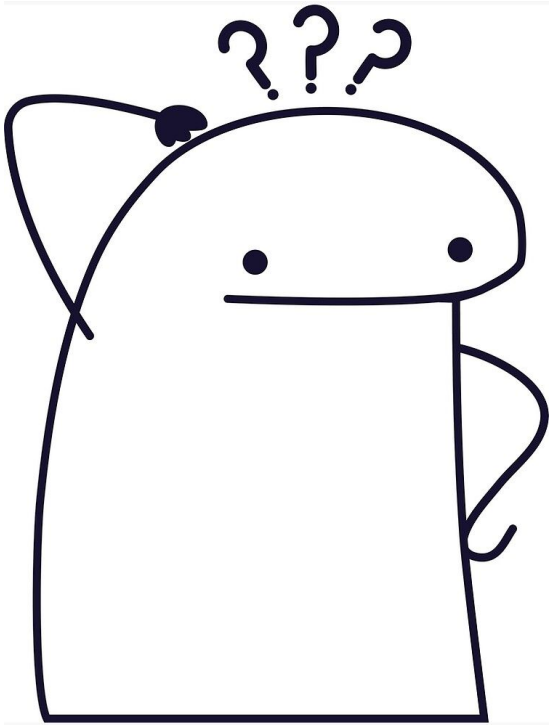


The delayed_job jobs table

```
create_table :delayed_jobs, :force => true do |table|
  table.integer  :priority, :default => 0      # Allows some jobs to jump to the front of the queue
  table.integer  :attempts, :default => 0      # Provides for retries, but still fail eventually.
  table.text     :handler                      # YAML-encoded string of the object that will do work
  table.text     :last_error                   # reason for last failure (See Note below)
  table.datetime :run_at                       # When to run. Could be Time.zone.now for immediately,
or sometime in the future.
  table.datetime :locked_at                    # Set when a client is working on this object
  table.datetime :failed_at                    # Set when all retries have failed (actually, by
default, the record is deleted instead)
  table.string   :locked_by                    # Who is working on this object (if locked)
  table.string   :queue                        # The name of the queue this job is in
  table.timestamps
end
```

How does it work?

- Additionally we need to tell ActiveJob to use delayed_job as queue adapter by setting the following config in the application.rb file: config.active_job.queue_adapter = :delayed_job
- After doing all this, the delayed_job worker can start working off the enqueued jobs



Use it!

- First off, run jobs:work to start working off the delayed jobs
- Call `.delay.method(params)` on any object and it will be processed in the background.

```
# without delayed_job
@user.activate!(@device)

# with delayed_job
@user.delay.activate!(@device)
```

- Delay() takes these parameters
  - :priority (number) lower numbers run first
  - :run_at (Time) run the job at this time
  - :queue (String) Named queue to put the job in

Delayed Cron Job

- We can use the delayed_cron_job gem to enqueue jobs that we want to execute repeatedly
- As the name says, the gem allows us to define a job with an assigned cron expression, which is then used to repeatedly execute the job

```ruby
class NightlyUpdatePeopleDataPtimeJob < CronJob
  self.cron_expression = '0 3 * * *'

  def perform(**args)
    if Skills.use_ptime_sync?
      Ptime::PeopleEmployees.new.update_people_data(**args)
    end
  end
end
```

Additional Tips and Tricks

- Additionally you can create a (empty) delayed_job model to access it on the rails console
- Crontab.guru is a useful website to check if your Cron expression works as expected

Links

https://github.com/collectiveidea/delayed_job

https://github.com/collectiveidea/delayed_job_active_record

https://github.com/codez/delayed_cron_job

https://crontab.guru/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